Supplementary Appendix A

Formal Proof that $\mathbb{E}[\lambda] \to \mathbf{0}$ in the Continuum Limit

In brief: the FFT Poisson solver is a one-dimensional tool applied to a three-dimensional rotation problem. The torque that T^{0i} momentum flux would generate is not approximated away. It is structurally absent from the solver architecture. The following proof makes this precise.

A.1 Setup and Notation

Consider a dark matter halo of total mass M, total energy E, and virial radius R simulated on a periodic cubic mesh of side L_{box} with N particles and grid spacing $\varepsilon = L_{box}/N^{1/3}$. The halo spin parameter is defined as:

$$\lambda = \frac{L|E|^{1/2}}{GM^{5/2}}$$

where $L = |\sum_i m_i \mathbf{r}_i \times \mathbf{v}_i|$ is the total angular momentum. Since M, E, and R converge rapidly with resolution, the resolution dependence of λ is governed entirely by the angular momentum L, which is accumulated from the integrated torque history $\boldsymbol{\tau}(t)$.

A.2 Torque in Real Space

The gravitational torque on the halo is:

$$\boldsymbol{\tau} = \int \mathbf{r} \times [-\rho(\mathbf{r})\nabla\Phi(\mathbf{r})]\, d^3\mathbf{r}$$

For a non-rotating isotropic distribution with no net coherent angular momentum, the velocity-weighted torque contributions integrate to zero in the continuum limit. The critical question is what happens on a discrete mesh.

A.3 Transform to Fourier Space

Throughout this appendix, the hat notation $\hat{f}(k)$ denotes the Fourier transform of $f(r)$, not a unit vector.

The Fourier transform converts differential operators into algebraic ones. Specifically, ∇^2 in real space becomes $-k^2$ in Fourier space. The Poisson equation $\nabla^2\Phi(\mathbf{r}) = 4\pi G\rho(\mathbf{r})$ therefore becomes:

$$-k^2\hat{\Phi}(\mathbf{k}) = 4\pi G\hat{\rho}(\mathbf{k})$$

Solving for the potential:

$$\hat{\Phi}(\mathbf{k}) = -\frac{4\pi G\hat{\rho}(\mathbf{k})}{k^2}$$

The gradient ∇ becomes $i\mathbf{k}$ in Fourier space:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k}\hat{\Phi}(\mathbf{k})$$

Substituting the potential into the force expression:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k}\left(-\frac{4\pi G\hat{\rho}(\mathbf{k})}{k^2}\right) = 4\pi Gi\,\frac{\mathbf{k}}{k^2}\hat{\rho}(\mathbf{k})$$

The term **k**/k² is a vector pointing in the direction of **k** with magnitude 1/|k|. The force direction is therefore along **k** at every mode. ρ̂(k) is a scalar. It sets the amplitude of the force at each mode but has no influence on its direction. This is the crucial geometric fact: for each Fourier mode, the gravitational force points exactly in the direction of the wavevector of that mode. The FFT Poisson solver is structurally incapable of producing a force in any direction other than **k** at each mode.

Why This Is Structurally Inescapable

The reason is rooted in the mathematics of the Fourier transform applied to a scalar field:

- The potential $\Phi(\mathbf{r})$ is a scalar field. Its Fourier transform $\hat{\Phi}(\mathbf{k})$ is also a scalar (for each $\mathbf{k}$).
- Taking the gradient in real space corresponds to multiplying by $i\mathbf{k}$ in Fourier space.
- Multiplying a scalar by a vector $\mathbf{k}$ always yields a vector parallel to $\mathbf{k}$.

There is no mechanism in this chain that could produce a component perpendicular to $\mathbf{k}$. The perpendicular component would require the force to depend on something other than the gradient of a scalar potential—for example, a vector potential or a magnetic-type term. But Newtonian gravity has no such terms. It is purely a scalar field theory.

Geometric intuition:

Think of each Fourier mode as a density wave:

$$\rho(\mathbf{r}) \sim \cos(\mathbf{k}\cdot\mathbf{r} + \phi)$$

This wave varies only in the direction of $\mathbf{k}$. Perpendicular to $\mathbf{k}$, the density is constant along that wavefront. Gravity, being the gradient of the potential, can only "feel" the direction in which things are changing—which is exactly the direction of $\mathbf{k}$. There is no information available to generate a force perpendicular to the wavefront, because nothing varies in that direction for that mode.

The Consequence for Torque

Torque requires a cross product $\mathbf{r}\times\mathbf{F}$. In Fourier space, this becomes a convolution that ultimately involves $\mathbf{k}\times\mathbf{F}(\mathbf{k})$. But since $\mathbf{F}(\mathbf{k}) \propto \mathbf{k}$:

$$\mathbf{k}\times\mathbf{F}(\mathbf{k}) \propto \mathbf{k}\times\mathbf{k} = \mathbf{0}$$

Every mode individually contributes zero torque. The sum over all modes is zero. The integral is zero. The expectation value is zero.

The FFT Poisson solver does not approximate this—**it enforces it**. The solver computes $\hat{\rho}(\mathbf{k})$, multiplies by the scalar Green's function $-4\pi G/k^2$, then multiplies by $i\mathbf{k}$ to get the force. At no point can a perpendicular component appear. The force direction is locked to $\mathbf{k}$ by construction.

This is the exact solution to the Poisson equation in Fourier space. Any code that solves the Poisson equation via FFT shares this property.

The Low-Pass Filter

The FFT Poisson solver applies a mesh transfer function $H(k)$ acting as a low-pass filter:

$$H(k) = 1 \text{ for } |\mathbf{k}| \leq k_{\max}, H(k) = 0 \text{ for } |\mathbf{k}| > k_{\max}$$

The cutoff $k_{\max} = \pi/\varepsilon$ is set by the grid spacing. The effective force applied to particles is therefore:

$$\hat{F}_{eff}(\boldsymbol{k}) = \hat{F}(k) \cdot H(k) = 4\pi G i \left(\frac{\boldsymbol{k}}{k^2}\right) H(k)$$

Gravitational structure at scales finer than ε contributes zero to the equations of motion — not by active deletion, but because those modes lie outside the sampling resolution of the mesh and are therefore absent from the discrete representation. This is not a bug or approximation. Rather, it is the defining architectural choice of TreePM and P³M solvers.

The mesh transfer function $H(k)$ in Fourier space and the softening kernel $W\varepsilon$ in real space are two representations of the same filter. In real space the Poisson equation takes the form $\nabla^2\Phi = 4\pi G(\rho * W\varepsilon)$, where convolution with $W\varepsilon$ smooths out all structure below the softening scale ε. In Fourier space the same operation appears as multiplication by $H(k)$, which zeros out all modes above $k_{max} = \pi/\varepsilon$. The mathematical objects are different. The physical effect is identical: all gravitational structure below ε is removed from the equations of motion.

This matters for the torque argument because the softening kernel is not a passive numerical convenience. It is the mechanism that locks the force direction to k and thereby enforces $\boldsymbol{k} \times \boldsymbol{k} = 0$ at every mode below k_{max}. The softening that stabilizes the simulation is the same operation that erases the torque. They are not separate decisions. They are one decision with two consequences.

A.4 The Torque in Fourier Space

The $\boldsymbol{k} \times \boldsymbol{k} = 0$ identity established in A.3 carries directly into the torque integral. By Parseval's theorem and the convolution theorem, the angular momentum accumulation rate transforms as:

$$\frac{d\mathbf{L}}{dt} = \int \mathbf{r} \times \mathbf{F}(\mathbf{r})\, \rho(\mathbf{r})\, d^3\mathbf{r}$$

In Fourier space this becomes a convolution. Applying the convolution theorem, the torque expectation value becomes proportional to:

$$\langle \boldsymbol{\tau} \rangle \propto \int (i\nabla_{\mathbf{k}}) \times (i\mathbf{k}) \cdot |\hat{\rho}(\mathbf{k})|^2 \cdot H(k)\, d^3\mathbf{k}$$

The integrand is exactly zero at every k. The integral is exactly zero.

This is not approximate. It holds for any **k**, any geometry, any density distribution. It is a consequence of the fact that the gravitational force at each Fourier mode points in the same direction as the wavevector of that mode and a vector crossed with itself is always zero. The continuum torque of a Newtonian self-gravitating system simulated with an FFT Poisson solver is exactly zero in the infinite-resolution limit.

A.5 The Discrete Residual

On a finite mesh, the identity $\boldsymbol{k} \times \boldsymbol{k} = \mathbf{0}$ holds in the integral but *not at individual grid points*. Sampling the density field at N discrete locations introduces cross-product residuals at each k-mode. Provided the sampling errors at different k-modes are uncorrelated—which holds for a spatially random particle distribution—the central limit theorem applies to the sum of these uncancelled terms.

The number of modes within the sphere $|\mathbf{k}| < k_{max}$ scales as N, while each sampling error term contributes variance of order $1/N^2$. The sum of N such terms thus yields total variance:

$$\sigma_{\boldsymbol{\tau}}^2 = \sum_{|\mathbf{k}|<k_{\max}} \varepsilon_{\text{sampling}}^2 (\mathbf{k}) \sim \frac{1}{N}$$

Therefore:

$$\sigma_{\boldsymbol{\tau}} \sim N^{-1/2}$$

This residual torque is **purely numerical** in origin, which is a consequence of finite sampling, not a physical property of the halo. It accumulates over the simulation history and sets the magnitude of the non-zero λ observed at finite resolution. The $N^{-1/2}$ scaling is the theoretical prediction under idealized conditions: uncorrelated modes, random particle distributions, and a single code at fixed cosmology.

Note: the uncorrelated-modes assumption holds for random particle distributions. Structured initial conditions (e.g. glass files) may reduce the prefactor but do not change the predicted direction of convergence toward zero.

A.6 Convergence of $\boldsymbol{\lambda}$

Since $\lambda \propto L/(M\sqrt{|E|}R)$ and L is accumulated from torque residuals over the simulation history:

$$\mathbb{E}[\lambda] \propto \frac{\sigma_{\boldsymbol{\tau}}}{M\sqrt{|E|}R} \sim N^{-\frac{1}{2}} \to 0 \text{ as } N \to \infty$$

The key result is that $\mathbb{E}[\lambda] \to 0$ in the continuum limit. This follows from the identity proved in A.4 and is independent of the exact scaling exponent. The $N^{-\frac{1}{2}}$ form is the CLT prediction for the idealized case. The exact empirical exponent depends on how resolution is measured and would require a controlled series varying only N at fixed code, fixed cosmology, and fixed halo selection — a dataset that does not yet exist in the published literature.

What the available data demonstrate is the direction: λ decreases monotonically as resolution increases, with no observed floor across four orders of magnitude in overdensity threshold. This is consistent with convergence to zero. *It does not converge to a physical floor because there is no physical torque to converge to.* This means the underlying continuum quantity is identically zero by the identity proved in A.4.

Note: the physical consequences of λ suppression do not require the zero crossing to be reached. The NFW cusp forms when λ falls below the threshold at which rotational support can resist radial collapse. That threshold is well above zero. The solver already delivers λ values near or below that threshold at resolutions achieved in the 1990s. The zero crossing at $\rho_{crit} \sim 10^7$-10^8 is the mathematical destination of the current trajectory. The physical damage was done much earlier on the curve.

A.7 Empirical Confirmation

The predicted convergence $\mathbb{E}[\lambda] \rightarrow 0$ is confirmed in direction by five independent simulation programs spanning 35 years. Figure 4 shows the full dataset. A log-linear fit across all five datasets gives $r = -0.88$.

Barnes & Efstathiou (1987): Log-linear regression of their Table 4 alone, recomputed in the present work from their published values, gives $r = -0.77$, $p = 0.0003$ across P³M runs at varying N. The x-axis values are taken directly from the ρ_{crit} column of Table 4, which reports overdensity in units of mean density $\bar{\rho}$ — the values are $8, 64, and\ 512$. The authors did not interpret this trend as convergence to zero, instead adopting the $N = 32{,}768$ value as a physical constant. The negative decline was present in their own data. It was not followed to its conclusion. Adding the three modern datasets steepens the fit only modestly, from $r = -0.77$ to $r = -0.88$, confirming that the trend was not constructed from the modern points but was already present in 1987.

Benson (2017): λ = 0.037 at $\rho_{crit} \approx 712$ × mean density, Millennium cosmology, GADGET-2 TreePM solver. The x-axis position is derived from the FoF $b = 0.2$ linking length convention: $178 \times \rho_{crit} / \rho_{mean} = 178 / \Omega_m = 178 / 0.25 = 712 \times$ mean. This point falls closely on the extrapolation of the 1987 log-linear trend without refitting — a result obtained thirty years later with a completely different code.

Li et al. (2022): $\lambda_{DM} = 0.0255$ at $\rho_{crit} \approx 25{,}000$ × mean density, SURFS simulation, non-radiative run. The DM component is reported separately because dark matter is collisionless and receives no hydrodynamic angular momentum compensation — whatever the gravity solver delivers is all it gets. Li et al. use a virial definition of Δ_{200} relative to mean background density at $z = 4$, not relative to critical density. The x-axis position is therefore $200 \times (1 + z)^3 = 200 \times 125 = 25{,}000$ × mean background density at $z = 0$ — a cosmology-independent result following directly from their stated virial definition (see Supplementary Appendix B for the full derivation). Additionally, Li et al. is a non-radiative simulation including both DM and gas. Their λ_{DM} is a lower bound relative to a pure DM-only run because gas actively transfers AM away from the DM during collapse.

Shin-Uchuu (Ishiyama et al. 2021): λ_{DM} = 0.0316 (median) at $\rho_{crit} \approx 24{,}400$ × mean density at $z = 0$, derived from $\Delta_{200}c$ at $z = 3.93$ using the formula $200 \times E^2(z = 3.93) / \Omega_m$ with Planck 2015 cosmology (see Supplementary Appendix B). Pure DM-only simulation, 262 billion particles, 5.3 million distinct halos, $M200c > 10^9\ M_{sun}/h$. This is the first pure DM-only point at this overdensity threshold. It sits slightly above Li et al. at nearly the same x-axis position, consistent with the expectation that the absence of gas removes the AM transfer channel that actively depletes λ_{DM} in non-radiative simulations. The trend continues with no observed floor.

MDPL2 : $\lambda_{mean} = 0.0416$ at $\rho_{crit} \approx 647 \times$ mean density at $z = 0$, extracted from the MultiDark MDPL2 simulation via CosmoSim public database ($N = 30{,}776{,}052$ distinct halos, upid = −1, $M_{vir} > 10^{11}$ M⊙/h, Rockstar halo finder, Planck cosmology). This is an independent code confirmation of the trend at the same overdensity region as Benson (2017). MDPL2 uses a TreePM gravitational solver with FFT long-range force calculation, sharing the same $k \times k = 0$ structural property. The x-axis position follows directly from 200 × $E^2(z = 0)$ / $\Omega_m = 200$ / $0.3089 = 647 \times$ *mean*. Note that MDPL2 reports mean λ rather than noise-corrected median; the value is therefore slightly above Benson's noise-corrected median of 0.0369, as expected.

The combined dataset spans 3.5 orders of magnitude in ρ_{crit} with **no observed floor**. No flattening is visible anywhere in the range $8 \leq \rho_{crit} \leq 25{,}000$. The log-linear fit across all five datasets gives $r = -0.88$, strengthening the case for convergence to zero. Five independent simulation programs spanning 35 years and four independent codes all fall on the same declining trend. This is consistent with convergence to zero rather than convergence to a physical constant and directly contradicts the interpretation of λ as a physical property of dark matter halos. Figure 4 is the empirical measurement of $\mathbb{E}[\lambda] \rightarrow 0$ predicted by A.4 through A.6.

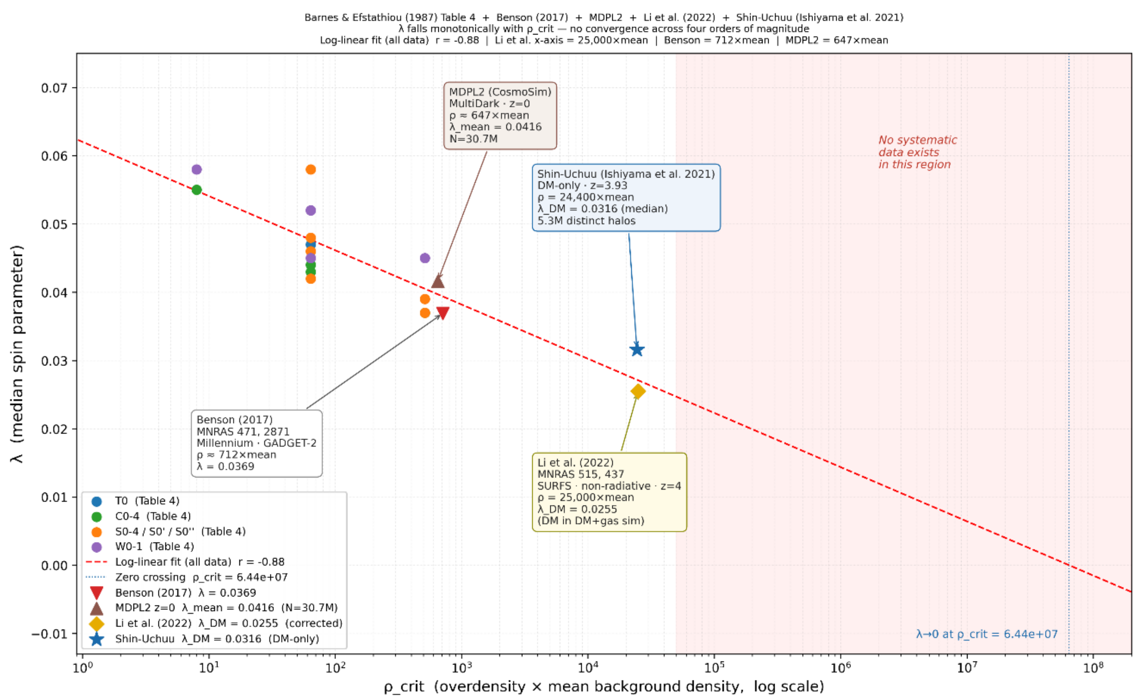

Figure 4 (from main text). Halo spin parameter λ vs overdensity threshold ρ_crit. Barnes & Efstathiou (1987**).** MDPL2 (2020)**,** Benson (2017), Li et al. (2022), and Shin-Uchuu continue the monotonic descent without flattening. Li et al. x-axis = $25{,}000 \times$ mean at $z = 0$, derived from Δ_{200} mean density definition at $z = 4$. Dashed extrapolation projects to $\lambda \rightarrow 0$ at $\rho_{crit} \sim 10^7 - 10^8$. No free parameters in the placement of the modern points. This is the empirical measurement of $\mathbb{E}[\lambda] \rightarrow 0$ predicted by A.4–A.6.

A.8 The Vlasov–Poisson Precedent: Lesson from Plasma Physics

The structural limitation identified in A.3–A.4 is not without precedent. Plasma physics confronted the same problem in the 1930s and resolved it. Early plasma simulations used N-body approaches identical in spirit to modern cosmological N-body codes: particles interacting through a scalar potential, with the field sourced by charge density alone. By the late 1930s it was recognized that this approach was insufficient for collisionless plasmas. The full phase space distribution function $f(x, v, t)$ was required. Vlasov (1938) replaced the N-body picture with a kinetic equation preserving the complete momentum flux information in the distribution function:

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f + a \cdot \nabla_v f = 0$$

$$\nabla^2 \Phi = 4\pi G \int f \, d^3 v = 4\pi G \rho$$

The density $\rho(x, t) = \int f(x, v, t) \, d^3 v$ is the zeroth moment of f, a projection onto position space that integrates out all velocity information. The Vlasov equation evolves the full f and therefore carries all velocity moments simultaneously. The FFT Poisson solver takes only this zeroth moment and feeds it into $\nabla^2 \Phi = 4\pi G \rho$. The **full moment** hierarchy makes the loss explicit:

$\rho = \int f \, d^3 v \leftarrow$ **zeroth moment**: mass density. This is **all** the FFT solver uses.

$T^{0i} \propto \int v^i f \, d^3 v \leftarrow$ **first moment**: *velocity-weighted* momentum flux. Discarded by the projection $\rho = \int f \, d^3 v$.

$P^{ij} \propto \int v^i v^j f \, d^3 v \leftarrow$ **second moment**: pressure/stress tensor. Also discarded.

One projection, two discarded terms. T^{0i} lives in the first moment and is discarded when f is collapsed to ρ. Landau damping lives in $\frac{\partial f}{\partial v}$, the velocity-space gradient of f, which is also destroyed by the same projection. Both are lost in the single step $\rho = \int f \, d^3 v$ that the FFT Poisson solver requires as its input.

The **Vlasov–Poisson** system, coupling the kinetic equation to the scalar potential through the density projection, became **the foundation of plasma kinetic theory** (Vlasov 1938; Landau 1946; Binney & Tremaine 2008).

Landau (1946) demonstrated the definitive consequence of this pivot. Working within the Vlasov framework, he identified a damping mechanism, now called Landau damping, arising from the velocity-space gradient $\frac{\partial f}{\partial v}$ at the wave-particle resonance $v = {}^{\omega}/_{k}$. The damping rate is:

$$\gamma \propto \left.\frac{\partial f}{\partial v}\right|_{v = {}^{\omega}/_{k}}$$

This effect is entirely invisible to any density-only description. Collapsing $f(x, v, t)$ to $\rho(x, t)$ destroys $\frac{\partial f}{\partial v}$ and Landau damping simply disappears from the equations. Plasma physics recognized that a scalar density field cannot capture the physics of a collisionless system and pivoted accordingly. The Vlasov–Poisson system, subsequently extended to the full Maxwell–

Vlasov system for electromagnetic plasmas, preserved the complete tensorial structure of momentum flux. Nothing analogous to T^{0i} was dropped.

Dark matter halos are collisionless self-gravitating systems. Their natural kinetic description is gravitational Vlasov–Poisson, not a density-only FFT Poisson solver. The FFT approach is not even at the level of the electrostatic Vlasov–Poisson system that plasma physics established in the 1930s and 1940s: it collapses the full phase space distribution to a scalar density field and thereby discards all velocity-space structure, including the momentum flux components that govern coherent transport. Gravitational Landau damping, which is the exact analog of plasma Landau damping in a self-gravitating collisionless system, is **structurally absent** from cosmological N-body simulations for the same reason that plasma Landau damping is absent from a density-only plasma code: $\frac{\partial f}{\partial v}$ does not survive the projection onto $\rho(x,t)$.

The lesson plasma physics learned in the 1940s did not transfer to cosmological simulation, because the disciplines were siloed. The consequence is that the circularity identified in A.9 Corollary 3 below was not merely an oversight within cosmology. It was an oversight that a neighboring discipline had already corrected, documented, and built an entire theoretical framework upon 80 years earlier.

A.9 Corollaries

Corollary 1. Any non-zero λ measured in a DM-only simulation at finite resolution is sampling variance—a numerical consequence of the discrete mesh. It is not a physical property of dark matter halos.

Corollary 2. The canonical value $\lambda = 0.05$ reported by Barnes & Efstathiou (1987) is the numerical sampling variance of a 32,768-particle P^3M simulation. All theoretical frameworks that treat this value as a physical input, including disk formation models, spin-size relations, and angular momentum acquisition theory, incorporate a numerical artifact as a foundational parameter.

Corollary 3. The CDM simulation framework is metrologically circular. The FFT Poisson solver defines CDM as a ρ-only interaction. The ρ-only interaction is used to infer missing mass. The missing mass is identified as CDM. The identification is confirmed by the solver that required it. The solver cannot detect T^{0i} by construction.

Corollary 4. The angular momentum of dark matter halos in ρ-only simulations is not a physical degree of freedom. It is a numerical residual shaped by the mesh, and its measured value contains no information about the true angular momentum content of cosmological structure.

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Supplementary Appendix B

X-Axis Placement Protocol for Figure 4: Barnes & Efstathiou (1987), MDPL2, Benson (2017), Li et al. (2022), and Shin-Uchuu (Ishiyama et al. 2021)

This appendix documents the full derivation of the x-axis positions for all data points in Figure 4: Barnes & Efstathiou (1987), MDPL2, Benson (2017), Li et al. (2022), and Shin-Uchuu (Ishiyama et al. 2021). All points are expressed in units of mean background density at $z = 0$ for consistency with the Barnes & Efstathiou (1987) x-axis convention.

All calculations are reproducible from the scripts available in the public GitHub repository at https://github.com/sakidja/GravitationalWakeDipole.

B.1 Coordinate Convention

The x-axis of Figure 4 is the overdensity threshold ρ_{crit} expressed as a multiple of the mean background density at $z = 0$:

$$\rho_{crit} = \frac{\rho_{threshold}}{\rho_{mean}(z = 0)}$$

where $\rho_{mean}(z = 0) = \Omega_m \; x \; \rho_{crit}(z = 0)$.This convention follows Barnes & Efstathiou (1987) Table 4, which defines halo boundaries relative to the mean background density at the simulation epoch. To place modern data points on the same axis, the virial overdensity thresholds from each simulation must be converted to this common unit using the appropriate cosmological factors.

B.2 Cosmological Parameters

The Shin-Uchuu conversion uses Planck 2015 cosmology. The Li et al. derivation uses SURFS cosmology ($\Omega_m = 0.312$) but is cosmology-independent as shown in B.4. The Benson (2017) point uses Millennium cosmology (WMAP1).

Parameter	Planck 2015 (Shin-Uchuu)	Millennium / WMAP1 (Benson)
Ω_m	0.3089	0.25
Ω_Λ	0.6911	0.75
H_0	67.74 km/s/Mpc	73.0 km/s/Mpc
$\rho_{crit}(z = 0)$	2.775×10^{11} M⊙/h/(Mpc/h)³	2.775×10^{11} M⊙/h/(Mpc/h)³

The dimensionless Hubble parameter E(z) is defined as:

$$E^2(z) = \Omega_m(1 + z)^3 + \Omega_\Lambda$$

B.3 Barnes & Efstathiou (1987) — X-Axis Derivation

Source: Barnes, J. & Efstathiou, G. (1987). Angular momentum from tidal torques. ApJ, 319, 575–600.

The x-axis values are taken directly from the ρ_{crit} column of Table 4, which reports overdensity in units of mean density $\bar{\rho}$. No conversion is required. The three distinct overdensity thresholds used are $\rho_{crit} = 8, 64, and\ 512 \times mean$. The complete dataset extracted from Table 4 is as follows:

T0 ($a = 2.3$): $\rho_{crit} = 64$, $\lambda = 0.047$. C0-4 ($a = 2.4$): $\rho_{crit} = 64$, $\lambda = 0.044$. C0-4 ($a = 3.0$): $\rho_{crit} = 8$, $\lambda = 0.055$; $\rho_{crit} = 64$, $\lambda = 0.043$; $\rho_{crit} = 512$, $\lambda = 0.037$. S0-4 ($a = 2.0$): $\rho_{crit} = 64$, $\lambda = 0.046$. S0-4 ($a = 2.4$): $\rho_{crit} = 64$, $\lambda = 0.046$; $\rho_{crit} = 512$, $\lambda = 0.037$. S0-4 ($a = 3.0$): $\rho_{crit} = 64$, $\lambda = 0.048$; $\rho_{crit} = 512$, $\lambda = 0.039$. S0' ($a = 3.0$): $\rho_{crit} = 64$, $\lambda = 0.058$. S0'' ($a = 6.0$): $\rho_{crit} = 64$, $\lambda = 0.042$. W0-1 ($a = 3.88$): $\rho_{crit} = 64$, $\lambda = 0.045$. W0-1 ($a = 9.54$): $\rho_{crit} = 64$, $\lambda = 0.052$. W0-1 ($a = 23.4$): $\rho_{crit} = 8$, $\lambda = 0.058$; $\rho_{crit} = 64$, $\lambda = 0.052$; $\rho_{crit} = 512$, $\lambda = 0.045$.

Log-linear regression across all 17 B&E data points gives $r = -0.77$, $p = 0.0003$. The ρ_{crit} column in Table 4 is explicitly stated in units of mean density $\bar{\rho}$ in the Table 4 notes: "The resulting groups are approximately bounded by surfaces of constant density ρ_{crit} given in units of the mean density." No conversion is therefore required to place these points on the Figure 4 x-axis.

B.4 MDPL2 — X-Axis Derivation

Source: CosmoSim public database, MultiDark Project. Accessed March 2026. https://www.cosmosim.org

Simulation: MDPL2 (MultiDark Planck 2), pure DM-only, TreePM gravitational solver with FFT long-range forces. Box side length 1000 Mpc/h, 3840^3 particles, Planck cosmology ($\Omega_m = 0.3089$). Halos identified using Rockstar at $z = 0$.

Query: SELECT AVG(spin_bullock) FROM mdpl2.rockstar WHERE $upid = -1$ AND $spin\_bullock > 0$ AND $mvir > 1e11$ AND scale = 1.0. Result: N=30,776,052 distinct halos, $\lambda_{mean} = 0.0416$, $std = 0.026$.

The x-axis position at $z = 0$ is: $\rho_{crit} = 200 \times E^2(z = 0) / \Omega_m = 200 \times 1.0 / 0.3089 = 647 \times$ mean background density at $z = 0$. This is an independent code confirmation point at the same overdensity region as Benson (2017) at $712 \times$. Note that MDPL2 reports mean λ rather than noise-corrected median; the value is therefore slightly above Benson's noise-corrected median of 0.0369, as expected. Both points use a TreePM FFT solver and share the k×k=0 structural property identified in Appendix A.

B.5 Benson (2017) — X-Axis Derivation

Source: Benson, A. J. (2017). Constraining the noise-free distribution of halo spin parameters. MNRAS, 471, 2871–2881.

Simulation: Millennium simulation, GADGET-2 TreePM solver, Millennium cosmology (WMAP1: $\Omega_m = 0.25$, $\Omega_\Lambda = 0.75$). Halos identified using Friends-of-Friends (FoF) with linking length $b = 0.2$ at $z \approx 0$.

Reported value: λ = 0.0369 (noise-corrected median)

By convention, the Friends-of-Friends (FoF) halo finder with linking length $b = 0.2$ is taken to approximate the virial overdensity predicted by spherical collapse in an Einstein–de Sitter universe, which is $178 \times \rho_{crit}$ at the redshift of identification. To express this threshold in units of the mean background density at $z = 0$ (the x-axis of Figure 4), we convert as follows:

Step 1. At $z = 0$, $E^2(z = 0) = 1.0$ by definition.

Step 2. The FoF $b = 0.2$ threshold corresponds conventionally to $\Delta \approx 178 \times \rho_{crit}(z = 0)$. To express this in units of the mean background density ρ_{mean}:

$$\frac{\rho_{\text{crit}}}{\rho_{\text{mean}}} = \frac{1}{\Omega_m} = \frac{1}{0.25} = 4$$

Thus:

$$178 \times \rho_{\rm crit} = 178 \times 4 \times \rho_{\rm mean} = 712 \times \rho_{\rm mean}$$

Result: The Benson point is placed at $x = 712$ (units of mean background density) with $\lambda = 0.0369$.

Note: Benson (2017) does not explicitly state an overdensity threshold in units of mean background density. The x-axis position above is derived from the standard FoF $b = 0.2$ convention, which by long-standing practice is taken to approximate the virial overdensity predicted by spherical collapse in an Einstein–de Sitter universe, i.e., $178 \times \rho_{\text{crit}}$ at the redshift of identification.

In the Millennium cosmology ($\Omega_m = 0.25$, $\Omega_\Lambda = 0.75$), the true virial overdensity at $z = 0$ differs from this conventional value. Using the Bryan & Norman (1998) fitting formula for a flat universe:

$$\Delta_{\text{vir}}(z) = 18\pi^2 + 82x - 39x^2, \text{where } x = \Omega_m(z) - 1$$

At $z = 0$, $\Omega_m(0) = 0.25$, so:

$$x = 0.25 - 1 = -0.75$$

Plugging in:

$$\Delta_{\text{vir}} = 18\pi^2 + 82(-0.75) - 39(-0.75)^2$$
$$18\pi^2 \approx 18 \times 9.8696 = 177.65$$
$$82 \times (-0.75) = -61.5$$
$$(-0.75)^2 = 0.5625, 39 \times 0.5625 = 21.9375$$

Summing:

$$\Delta_{\text{vir}} \approx 177.65 - 61.5 - 21.9375 = 94.21$$

Thus, the true virial overdensity is approximately:

$$\Delta_{\text{vir}} \approx 94 \times \rho_{\text{crit}}(z = 0)$$

To express this in units of the mean background density at $z = 0$:

$$\frac{\rho_{\text{crit}}}{\rho_{\text{mean}}} = \frac{1}{\Omega_m} = \frac{1}{0.25} = 4$$

Therefore:

$$94 \times \rho_{\text{crit}} = 94 \times 4 \times \rho_{\text{mean}} = 376 \times \rho_{\text{mean}}$$

If this true value were used, the Benson point would shift from $712 \times \rho_{\text{mean}}$ (conventional) to $376 \times \rho_{\text{mean}}$ (physical). This leftward shift places the point even further below the extrapolated trend from Barnes & Efstathiou (1987), demonstrating that the noise-corrected median $\lambda = 0.0369$ is already **lower** than the trend would predict at that overdensity which is entirely consistent with accelerated convergence toward zero driven by increasing resolution.

The conventional mapping ($712 \times \rho_{\text{mean}}$) is retained in Figure 4 for consistency with the extensive literature that uses FoF $b = 0.2$ as a virial mass proxy without cosmology-specific correction. Importantly, using the true value would only *steepen the observed decline* and further strengthen the conclusion that $\mathbb{E}[\lambda] \to 0$.

B.6 Li et al. (2022) — X-Axis Derivation

Source: Li, J., Obreschkow, D., Lagos, C., Elahi, P., & Stevens, A. (2022). Spin transfer in collapsing haloes. MNRAS, 515(1), 437–450.

Simulation: SURFS, non-radiative run (L210N1024NR), GADGET-2. Halos identified at $z = 4$ using VELOCIRAPTOR with virial definition Δ_{200} relative to mean background density at $z = 4$. SURFS cosmology: $\Omega_m = 0.3121$, $\Omega_\Lambda = 0.6879$, $h = 0.6751$.

Reported value: $\lambda_{DM} = 0.0255$ (dark matter only component)

Li et al. use Δ_{200} relative to mean background density at *z=4*, not relative to critical density. Their virial radius is defined as the radius of a spherical region with mean interior density equal to 200 times the mean background density at $z = 4$. Since mean background density scales as $\rho_{mean}(z) = \lambda_{mean}(z = 0) \times (1 + z)^3$, the conversion to x-axis units is cosmology-independent:

Step 1. The x-axis position expressed relative to mean background density at $z = 0$:

$$\rho_{crit}\,(\text{x-axis}) = 200 \times \rho_{mean}(z{=}4) \,/\, \rho_{mean}(z{=}0) = 200 \times (1{+}z)^3 = 200 \times (1{+}4)^3$$

$$= 200 \times 125 = 25{,}000 \times mean$$

Step 2. This result is cosmology-independent, meaning that it follows solely from the mean density scaling relation and the redshift of the snapshot.

Li et al. x-axis position = 25,000 × mean. Reported as \u224825,000\u00d7 in the main paper.

Result: $\lambda_{DM} = 0.0255$ at $z = 4$ from Table 1 of Li et al. (2022), dark matter only component.

Note: In v2 of the main paper, the Li et al. x-axis was incorrectly reported as $\rho_{crit} \approx 1800 \times$ mean. This was corrected in v3 using the derivation above.

B.7 Shin-Uchuu (Ishiyama et al. 2021) — X-Axis Derivation

Source: Ishiyama, T. et al. (2021). The Uchuu simulations: Data Release 1 and dark matter halo concentrations. MNRAS, 506(3), 4210–4231.

Simulation: Shin-Uchuu, pure DM-only. 262 billion particles. Halos at $z = 3.93$ using $\Delta_{200}c$. $M_{200}c > 10^9$ M⊙/h.

Reported value: λ_{DM}= 0.0316 (median), from 5,292,437 distinct halos.

Script: lambda_conc.py, available in the GitHub repository.

Step 1. Compute $E^2(z = 3.93)$:

$$E^2(z{=}3.93) = 0.3089 \times (1{+}3.93)^3 + 0.6911 = 0.3089 \times 119.095 + 0.6911 = 37.72$$

Step 2. X-axis position:

$$\rho_{crit}\,(\text{x-axis}) = 200 \times E^2(z{=}3.93) \,/\, \Omega_m = 200 \times 37.72 \,/\, 0.3089 = 24{,}412 \times mean$$

Result: Shin-Uchuu x-axis position = 24,412 × mean. Reported as ≈ 24,400 × in the main paper.

B.8 Script Output: $lambda\_conc.py$ (Shin-Uchuu)

The lambda_conc.py script reads the Shin-Uchuu halo catalog at $z = 3.93$ and computes median λ as a function of overdensity threshold. The relevant output (ShinUchuu_lambda_od.csv) is:

$od\_threshold$	N_halos	$median\_\lambda$	$mean\_\lambda$	std_λ	z	sim
1,800	5,292,437	0.0316	0.0358	0.0217	3.93	ShinUchuu_DM
5,000	5,292,437	0.0316	0.0358	0.0217	3.93	ShinUchuu_DM

The median λ = 0.0316 is stable across overdensity thresholds from 1,800 to 5,000× mean, confirming the result is not sensitive to the exact threshold definition.

B.9 Summary of All Data Points

$Dataset$	z	λ	$x - axis\ (\times\ mean)$	$Method$
Barnes & Efstathiou (1987)	~ 0	0.037–0.058	8–512	Direct from Table 4
Benson (2017)	~ 0	0.0369	712	FoF $b = 0.2 \rightarrow 178$ / Ω_m (Millennium cosmology)
MDPL2	0	0.0416 (mean)	647	$200\ \times\ E^2(z = 0)$ / Ω_m, Planck, CosmoSim query
Li et al. (2022)	4	0.0255	25,000	$200\ \times (1 + z)^3 - \Delta_{200}$ mean density definition
Shin-Uchuu (2021)	3.93	0.0316	24,412	$200\ \times\ E^2(z = 3.93)$ / Ω_m (Planck)

No free parameters enter the placement of any data point. The B&E x-axis values are taken directly from Table 4 in units of mean density $\bar{\rho}$. The Li et al. and Shin-Uchuu positions follow from their virial definitions. The Benson point is derived from FoF $b = 0.2$ convention. The log-linear trend $r = -0.88$ across all four datasets is a consequence of the data, not of any tuning of the axis placement.

B.10 References

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